$$\begin{aligned}\mathcal{C}_{\mathrm{osc}}(\tau) &= \int \mathrm{d}\Delta\, J(\Delta) \tanh\left(\frac{\Delta}{2k_{\mathrm{B}}T}\right)\left\{\coth\left(\frac{\Delta}{2k_{\mathrm{B}}T}\right)\cos\left(\Delta\tau\right) - \mathrm{i}\sin\left(\Delta\tau\right)\right\} \\ &= \int \mathrm{d}\Delta\, J(\Delta)\left\{\cos\left(\Delta\tau\right) - \mathrm{i}\tanh\left(\frac{\Delta}{2k_{\mathrm{B}}T}\right)\sin\left(\Delta\tau\right)\right\}. \qquad (5.190)\end{aligned}$$